

DTSA 5749

Model and Error Analysis for Natural Language Processing

FORMAT	CATEGORY	CREDITS
Online (DTSA)	Elective	1
SECTION	BEST ROLE MATCH	AVG ALIGNMENT
Section 4 – Elective	AI Engineer (78.5 / 100)	56.6 / 100
SPECIALIZATION Natural Language Processing		

57

CURRENT AVG SCORE
across 6 job roles

92

POTENTIAL SCORE
with recommended additions

+50

MAX SCORE GAIN
best single-role improvement

CURRENT COURSE DESCRIPTION

Understanding the performance of natural language processing models goes beyond simply computing measures like accuracy. In this course we will learn methods for analyzing the strengths and weaknesses of NLP systems, both neural and non-neural. We will also learn about problematic biases in NLP data and systems. Methods covered include standard benchmarks, qualitative error analysis, confusion matrices, contrastive and diagnostic evaluation, and probing experiments.

INDUSTRY ALIGNMENT BY JOB ROLE

JOB ROLE	CURRENT SCORE	PROJECTED SCORE	POINT GAIN	PRIORITY
AI Engineer	78.5	100.0	+21.5	MEDIUM
Data Scientist	55.9	90.7	+34.7	HIGH
Machine Learning Engineer	55.5	86.1	+30.6	HIGH
MLOps Engineer	51.0	91.9	+41.0	HIGH
Applied Scientist	49.9	100.0	+50.1	HIGH
ML Engineer	48.6	83.2	+34.6	HIGH

RECOMMENDED TOPICS TO ADD

Keywords highlighted in gold (**example**) appear as gaps across 3 or more job roles.

System Design 1 of 6 roles

scale

Deep Learning 1 of 6 roles

training

Business Communication 1 of 6 roles

business

PRIORITY RECOMMENDATIONS FOR INSTRUCTORS

1 **LOW** **Optional: System Design** — would broaden appeal to one additional role. Candidate terms: scale.

2 **LOW** **Optional: Deep Learning** — would broaden appeal to one additional role. Candidate terms: training.

3 **LOW** **Optional: Business Communication** — would broaden appeal to one additional role. Candidate terms: business.

GAP KEYWORDS BY ROLE

AI Engineer

78.5 → 100.0 (+21.5 pts)

learning infrastructure software model problems technology detection design

Data Scientist

55.9 → 90.7 (+34.7 pts)

sets analytics social problems approaches

Machine Learning Engineer

55.5 → 86.1 (+30.6 pts)

sets learning analytics tools exploratory business mobile design

MLOps Engineer

51.0 → 91.9 (+41.0 pts)

infrastructure detection model text learning image training tools

Applied Scientist

49.9 → 100.0 (+50.1 pts)

biology lab discovery scientific computational research problems learning

ML Engineer

48.6 → 83.2 (+34.6 pts)

discovery learning model infrastructure problems impact scale building

Methodology: Scores (0–100) reflect TF-IDF cosine similarity between course catalog descriptions and a corpus of 100 LinkedIn data science job postings (collected June 2026, filtered to postings with ≥200 applicants). Gap keywords are terms that appear frequently in job postings for a given role but are absent or underrepresented in the current course description. "Projected score" shows the estimated alignment after integrating the top 5 suggested keywords. Analysis was conducted by the CU Boulder MSDS Curriculum-Industry Alignment project team.